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METHOD FOR REDUCING VCO COUPLING AND ELECTRONIC DEVICE SUPPORTING SAME

Abstract

According to one embodiment, an electronic device may be provided, the electronic device comprising an RF circuit and a communication processor operatively connected to the RF circuit, wherein the RF circuit comprises at least one reception circuit, at least one transmission circuit, a first VCO, a second VCO, and a signal detection block electrically connected to the first VCO, and the communication processor is configured to control the RF circuit to change the operating frequency of the second VCO on the basis of determining whether a signal output from the first VCO is an abnormal signal. Various other embodiments may be possible.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2023/020448 designating the United States, filed on Dec. 12, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0173038, filed on Dec. 12, 2022, and Korean Patent Application No. 10-2023-0005226, filed on Jan. 13, 2023, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

[0002] The disclosure relates to a method for reducing VCO coupling and an electronic device for supporting the same.

Background Art

[0003] As mobile communication technology evolves, multi-functional portable terminals are commonplace and, to meet increasing demand for radio traffic, vigorous efforts are underway to develop 5G communication systems. To achieve a higher data transmission rate, 5G communication systems are being implemented on ultra-high frequency bands as well as those used for 3G communication systems and long-term evolution (LTE) communication systems.

[0004] For example, to mitigate pathloss on the mmWave band and increase the reach of radio waves, the following techniques are taken into account for the 5G communication system: beamforming, massive multi-input multi-output (MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beamforming, and large scale antenna.

[0005] Wireless communication systems are evolving to support a higher data rate to meet the demand for steadily increasing radio data traffic. Electronic devices may transmit and receive signals having 4G frequencies, 5G sub6 frequencies, and 3 GHz to 5 GHz frequencies to enhance network access and data transmission rates.

[0006] A radio frequency front end (RFFE) and an RF circuit of an electronic device may include more components to support carrier aggregation (CA) and E-UTRA new radio dual-connectivity (ENDC). The space for disposing the components of the RFFE and the RF circuit is gradually decreasing due to an increase in the size of the battery of the electronic device.

DISCLOSURE OF INVENTION

Solution to Problems

[0007] According to an embodiment of the disclosure, an electronic device may comprise an RF circuit and a communication processor operatively connected to the RF circuit. The RF circuit of the electronic device may be configured to include at least one reception circuit, at least one transmission circuit, a first VCO, a second VCO, and a signal detection block electrically connected to the first VCO. The communication processor of the electronic device may be configured to control the RF circuit to change an operating frequency of the second VCO based on identifying whether a signal output from the first VCO is an abnormal signal.

[0008] According to an embodiment of the disclosure, a method for operating an electronic device

may comprise detecting or identifying a signal output from a first VCO included in an RF circuit of the electronic device through a signal detection block included in the RF circuit. The method for operating the electronic device may comprise identifying whether the detected or identified signal is an abnormal signal (e.g. the detected signal is identified as normal or abnormal) based on identifying a strength of the detected or identified signal. The method for operating the electronic device may comprise controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the detected or identified signal is an abnormal signal.

[0009] According to an embodiment of the disclosure, in a non-transitory storage medium storing instructions, the instructions may be configured to, when executed by at least one circuit of an electronic device, enable the electronic device to perform at least one operation. The at least one operation may include identifying a signal output from a first VCO included in an RF circuit of the electronic device through a signal detection block included in the RF circuit. The at least one operation may include identifying whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. The at least one operation may include controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the identified signal is an abnormal signal.

[0010] An embodiment of the disclosure is not limited to the foregoing objectives, and other objectives would readily be appreciated by a skilled artisan from the following detailed description taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

[0012] FIG. 2A is a block diagram illustrating an electronic device for supporting legacy network communication and 5G network communication according to an embodiment of the disclosure;

[0013] FIG. 2B is a block diagram illustrating an electronic device for supporting legacy network communication and 5G network communication according to an embodiment of the disclosure;

[0014] FIG. 3 is a block diagram illustrating an electronic device according to an embodiment of the disclosure;

[0015] FIG. 4 is a block diagram illustrating a RFIC according to an embodiment of the disclosure;

[0016] FIG. 5 is a view illustrating each component included in an RFIC according to an embodiment of the disclosure;

[0017] FIGS. 6A, 6B, and 6C are views illustrating an example of coupling between VCOs according to an embodiment of the disclosure;

[0018] FIG. 7 is a block diagram illustrating a signal detection block according to an embodiment of the disclosure;

[0019] FIG. 8 is a flowchart illustrating an operation of changing an operating frequency of a VCO of an electronic device according to an embodiment of the disclosure;

[0020] FIG. 9 is a flowchart illustrating an operation of changing an operating frequency of a VCO based on a strength of a signal received through a reception circuit by an electronic device according to an embodiment of the disclosure;

[0021] FIG. 10 is a flowchart illustrating an operation of changing an operating frequency of a VCO based on identifying a lookup table by an electronic device according to an embodiment of the disclosure; and

[0022] FIG. 11 is a flowchart illustrating an operation of changing a reference clock based on identifying a lookup table by an electronic device according to an embodiment of the disclosure.

[0023] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0024] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with at least one of an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In an embodiment, at least one (e.g., the connecting terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. According to an embodiment, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be integrated into a single component (e.g., the display module **160**).

[0025] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be configured to use lower power than the main processor **121** or to be specified for a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0026] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. The artificial intelligence model may be generated via machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent

neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0027] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0028] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0029] The input module **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, keys (e.g., buttons), or a digital pen (e.g., a stylus pen).

[0030] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0031] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display **160** may include a touch sensor configured to detect a touch, or a pressure sensor configured to measure the intensity of a force generated by the touch.

[0032] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0033] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0034] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0035] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0036] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may

include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0037] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0038] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0039] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0040] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via a first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0041] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0042] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module **197** may include one antenna including a radiator formed of a conductor or conductive pattern formed on a

substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., an antenna array). In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected from the plurality of antennas by, e.g., the communication module **190**. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **197**.

[0043] According to an embodiment, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0044] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0045] According to an embodiment, instructions or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. The external electronic devices **102** or **104** each may be a device of the same or a different type from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0046] FIG. 2A is a block diagram illustrating an electronic device for supporting legacy network communication and 5G network communication according to an embodiment. FIG. 2B is a block diagram illustrating an electronic device for supporting legacy network communication and 5G network communication according to an embodiment.

[0047] Referring to FIG. 2A, the electronic device **101** may include a first communication processor **212**, a second communication processor **214**, a first radio frequency integrated circuit (RFIC) **222**, a second RFIC **224**, a third RFIC **226**, a fourth RFIC **228**, a first radio frequency front end (RFFE) **232**, a second RFFE **234**, a first antenna module **242**, a second antenna module **244**, a third antenna module **246**, and antennas **248**. The electronic device **101** may further include a

processor **120** and a memory **130**. The second network **199** may include a first cellular network **292** and a second cellular network **294**. According to an embodiment, the electronic device **101** may further include at least one component among the components of FIG. **1**, and the second network **199** may further include at least one other network. According to an embodiment, the first communication processor **212**, the second communication processor **214**, the first RFIC **222**, the second RFIC **224**, the fourth RFIC **228**, the first RFFE **232**, and the second RFFE **234** may form at least part of the wireless communication module **192**. According to another embodiment, the fourth RFIC **228** may be omitted or be included as part of the third RFIC **226**.

[0048] The first communication processor **212** may establish a communication channel of a band that is to be used for wireless communication with the first cellular network **292** or may support legacy network communication via the established communication channel. According to an embodiment, the first cellular network may be a legacy network that includes second generation (2G), third generation (3G), fourth generation (4G), or long-term evolution (LTE) networks. The second communication processor **214** may establish a communication channel corresponding to a designated band (e.g., from about 6 GHz to about 60 GHz) among bands that are to be used for wireless communication with the second cellular network **294** or may support fifth generation (5G) network communication via the established communication channel. According to an embodiment, the second cellular network **294** may be a 5G network defined by the 3rd generation partnership project (3GPP). Additionally, according to an embodiment, the first CP **212** or the second CP **214** may establish a communication channel corresponding to another designated band (e.g., about 6 GHz or less) among the bands that are to be used for wireless communication with the second cellular network **294** or may support fifth generation (5G) network communication via the established communication channel.

[0049] The first communication processor **212** may perform data transmission/reception with the second communication processor **214**. For example, data classified as transmitted via the second cellular network **294** may be changed to be transmitted via the first cellular network **292**. In this case, the first communication processor **212** may receive transmission data from the second communication processor **214**. For example, the first communication processor **212** may transmit/receive data to/from the second communication processor **214** via an inter-processor interface **213**. The inter-processor interface **213** may be implemented as, e.g., universal asynchronous receiver/transmitter (UART) (e.g., high speed-UART (HS-UART)) or peripheral component interconnect bus express (PCIe) interface, but is not limited to a specific kind. The first communication processor **212** and the second communication processor **214** may exchange packet data information and control information using, e.g., a shared memory. The first communication processor **212** may transmit/receive various types of information, such as sensing information, information about output strength, and resource block (RB) allocation information, to/from the second communication processor **214**.

[0050] According to implementation, the first communication processor **212** may not be directly connected with the second communication processor **214**. In this case, the first communication processor **212** may transmit/receive data to/from the second communication processor **214** via a processor **120** (e.g., an application processor). For example, the first communication processor **212** and the second communication processor **214** may transmit/receive data to/from the processor **120** (e.g., an application processor) via an HS-UART interface or PCIe interface, but the kind of the interface is not limited thereto. The first communication processor **212** and the second communication processor **214** may exchange control information and packet data information with the processor **120** (e.g., an application processor) using a shared memory.

[0051] According to an embodiment, the first CP **212** and the second CP **214** may be implemented in a single chip or a single package. According to an embodiment, the first communication processor **212** or the second communication processor **214**, along with the processor **120**, an assistance processor **123**, or communication module **190**, may be formed in a single chip or single

package. For example, as shown in FIG. 2B, a communication processor **440** may support all of the functions for communication with the first cellular network **292** and the second cellular network **294**.

[0052] Upon transmission, the first RFIC **222** may convert a baseband signal generated by the first communication processor **212** into a radio frequency (RF) signal with a frequency ranging from about 700 MHz to about 3 GHz which is used by the first cellular network **292** (e.g., a legacy network). Upon receipt, the RF signal may be obtained from the first cellular network **292** (e.g., a legacy network) through an antenna (e.g., the first antenna module **242**) and be pre-processed via an RFFE (e.g., the first RFFE **232**). The first RFIC **222** may convert the pre-processed RF signal into a baseband signal that may be processed by the first CP **212**.

[0053] Upon transmission, the second RFIC **224** may convert the baseband signal generated by the first CP **212** or the second CP **214** into a Sub6-band (e.g., about 6 GHz or less) RF signal (hereinafter, “5G Sub6 RF signal”) that is used by the second cellular network **294** (e.g., a 5G network). Upon receipt, the 5G Sub6 RF signal may be obtained from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., the second antenna module **244**) and be pre-processed via an RFFE (e.g., the second RFFE **234**). The second RFIC **224** may convert the pre-processed 5G Sub6 RF signal into a baseband signal that may be processed by a corresponding processor of the first communication processor **212** and the second communication processor **214**.

[0054] The third RFIC **226** may convert the baseband signal generated by the second CP **214** into a 5G Above6 band (e.g., from about 6 GHz to about 60 GHz) RF signal (hereinafter, “5G Above6 RF signal”) that is to be used by the second cellular network **294** (e.g., a 5G network). Upon receipt, the 5G Above6 RF signal may be obtained from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., the antenna **248**) and be pre-processed via the third RFFE **236**. The third RFIC **226** may convert the pre-processed 5G Above6 RF signal into a baseband signal that may be processed by the second communication processor **214**. According to an embodiment, the third RFFE **236** may be formed as part of the third RFIC **226**.

[0055] According to an embodiment, the electronic device **101** may include the fourth RFIC **228** separately from, or as at least part of, the third RFIC **226**. In this case, the fourth RFIC **228** may convert the baseband signal generated by the second communication processor **214** into an intermediate frequency band (e.g., from about 9 GHz to about 11 GHz) RF signal (hereinafter, “IF signal”) and transfer the IF signal to the third RFIC **226**. The third RFIC **226** may convert the IF signal into a 5G Above6 RF signal. Upon receipt, the 5G Above6 RF signal may be received from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., the antenna **248**) and be converted into an IF signal by the third RFIC **226**. The fourth RFIC **228** may convert the IF signal into a baseband signal that may be processed by the second communication processor **214**.

[0056] According to an embodiment, the first RFIC **222** and the second RFIC **224** may be implemented as at least part of a single chip or single package. According to an embodiment, when the first RFIC **222** and the second RFIC **224** in FIG. 2A or 2B are implemented as a single chip or a single package, they may be implemented as an integrated RFIC. In this case, the integrated RFIC is connected to the first RFFE **232** and the second RFFE **234** to convert a baseband signal into a signal of a band supported by the first RFFE **232** and/or the second RFFE **234**, and may transmit the converted signal to one of the first RFFE **232** and the second RFFE **234**. According to an embodiment, the first RFFE **232** and the second RFFE **234** may be implemented as at least part of a single chip or single package. According to an embodiment, at least one of the first antenna module **242** or the second antenna module **244** may be omitted or be combined with another antenna module to process multi-band RF signals.

[0057] According to an embodiment, the third RFIC **226** and the antenna **248** may be disposed on the same substrate to form the third antenna module **246**. For example, the wireless communication module **192** or the processor **120** may be disposed on a first substrate (e.g., a main printed circuit board (PCB)). In this case, the third RFIC **226** and the antenna **248**, respectively, may be disposed

on one area (e.g., the bottom) and another (e.g., the top) of a second substrate (e.g., a sub PCB) which is provided separately from the first substrate, forming the third antenna module **246**.

Placing the third RFIC **226** and the antenna **248** on the same substrate may shorten the length of the transmission line therebetween. This may reduce a loss (e.g., attenuation) of high-frequency band (e.g., from about 6G Hz to about 60 GHz) signal used for 5G network communication due to the transmission line. Thus, the electronic device **101** may enhance the communication quality with the second cellular network **294** (e.g., a 5G network).

[0058] According to an embodiment, the antenna **248** may be formed as an antenna array which includes a plurality of antenna elements available for beamforming. In this case, the third RFIC **226** may include a plurality of phase shifters **238** corresponding to the plurality of antenna elements, as part of the third RFFE **236**. Upon transmission, the plurality of phase shifters **238** may change the phase of the 5G Above6 RF signal which is to be transmitted to the outside (e.g., a 5G network base station) of the electronic device **101** via their respective corresponding antenna elements. Upon receipt, the plurality of phase shifters **238** may change the phase of the 5G Above6 RF signal received from the outside to the same or substantially the same phase via their respective corresponding antenna elements. This enables transmission or reception via beamforming between the electronic device **101** and the outside.

[0059] The second cellular network **294** (e.g., a 5G network) may be operated independently (e.g., as standalone (SA)) from, or in connection (e.g., as non-standalone (NSA)) with the first cellular network **292** (e.g., a legacy network). For example, the 5G network may have the access network (e.g., 5G radio access network (RAN) or next generation RAN (NG RAN)) but may not have the core network (e.g., next generation core (NGC)). In this case, the electronic device **101**, after accessing a 5G network access network, may access an external network (e.g., the Internet) under the control of the core network (e.g., the evolved packet core (EPC)) of the legacy network. Protocol information (e.g., LTE protocol information) for communication with the legacy network or protocol information (e.g., New Radio (NR) protocol information) for communication with the 5G network may be stored in the memory **230** and be accessed by other components (e.g., the processor **120**, the first communication processor **212**, or the second communication processor **214**).

[0060] FIG. **3** is a block diagram illustrating an electronic device according to an embodiment of the disclosure.

[0061] According to an embodiment, the electronic device **101** may include at least one antenna module **340**, at least one RFFE **330** electrically connected to the at least one antenna module **340**, at least one RFIC **320** electrically connected to the at least one RFFE **330**, and a communication processor **310** operatively connected to the at least one RFIC **320**.

[0062] In an embodiment, the antenna module **340** may be included in at least one of the antenna module **197** of FIG. **1**, the first antenna module **242**, the second antenna module **244**, or the third antenna module **246** of FIG. **2A**.

[0063] In an embodiment, the RFFE **330** may be included in at least one of the first RFFE **232**, the second RFFE **234**, or the third RFFE **236** of FIG. **2A**. The RFFE **330** may include at least one power amplification (PA) **331**, at least one low noise amplification (LNA) **333**, a duplexer **335** electrically connected to the PA **331** and the LNA **333**, and a coupler **337** electrically connected to the duplexer **335**. The PA **331** may amplify a signal output by the transmission circuit **321** of the RFIC **320**. The signal amplified by the PA **331** may be transmitted to the duplexer **335**. The LNA **333** may amplify the signal passing through the duplexer **335**. The signal amplified by the LNA **333** may be transmitted to the reception circuit **323** of the RFIC **320**. The signal amplified by the PA **331** may pass through the duplexer **335**. The signal amplified by the PA **331** may be transmitted to the antenna module **340** through the duplexer **335** and the coupler **337**. The signal received by the antenna module **340** may pass through the duplexer **335**. The received signal may be transmitted to the LNA **333** through the coupler **337** and the duplexer **335**. When the signal

amplified by the PA 331 passes through the coupler 337, a feedback signal may be generated. The feedback signal generated by the coupler 337 may be transmitted to the feedback circuit 325 of the RFIC 320.

[0064] In an embodiment, the RFIC 320 may be included in at least one of the first RFIC 222, the second RFIC 224, the third RFIC 226, or the fourth RFIC 228 of FIG. 2A. The RFIC 320 may be configured to include at least one transmission circuit 321, at least one reception circuit 323, and at least one feedback circuit 325. The RFIC 320 may be electrically connected to the communication processor 310 through a data signal line 311 and a control signal line 313. The RFIC 320 may transmit and receive data to and from the communication processor 310 through the data signal line 311. The RFIC 320 may receive the control signal from the communication processor 310 through the control signal line 313. For example, the RFIC 320 may receive a control signal for controlling at least one of the transmission circuit 321, the reception circuit 323, or the feedback circuit 325 from the communication processor 310 through the control signal line 313. The transmission circuit 321 may generate an RF signal based on receiving a digital signal from the communication processor 310. The reception circuit 323 may generate a digital signal to be transmitted to the communication processor 310 based on receiving an RF signal through the antenna module 340. The feedback circuit 325 may generate a signal for detecting transmission power of the antenna module 340, based on the signal received from the coupler 337.

[0065] In an embodiment, the communication processor 310 may control an overall operation for reducing coupling between voltage controlled oscillators (hereinafter, referred to as “VCO”). In an embodiment, the communication processor 310 may be included in at least one of the processor 120 of FIG. 1, the first communication processor 212 of FIG. 2A, the second communication processor 214, or the integrated communication processor 260 of FIG. 2B. The operation performed by the communication processor 310 to reduce VCO coupling is described below with reference to FIGS. 4 to 11.

[0066] Although FIG. 3 illustrates that the electronic device 101 includes the communication processor 310, the RFIC 320, the RFFE 330, and/or the antenna module 340, the disclosure is not limited thereto. For example, the electronic device 101 may further include at least one component shown in FIG. 1.

[0067] FIG. 4 is a block diagram illustrating a RFIC according to an embodiment of the disclosure.

[0068] In an embodiment, the RFIC 320 (e.g., at least one of the first RFIC 222, the second RFIC 224, the third RFIC 226, the fourth RFIC 228, or the RFIC 320 of FIG. 3) may be configured to include at least one transmission circuit 321, at least one reception circuit 323, at least one transmission VCO 401 electrically connected to the at least one transmission circuit 321, at least one reception VCO 403 electrically connected to the at least one reception circuit 323, and a signal detection block 411 electrically connected to the reception VCO 403.

[0069] In an embodiment, the RFIC 320 may be configured to include one or more VCOs to transmit and receive signals based on one or more frequency combinations, but is not limited to that illustrated in FIG. 4. The electronic device 101 may support carrier aggregation (CA) or EN-DC based on operations of one or more VCOs. In an embodiment, one VCO may be configured to be electrically connected to eight reception circuits 323. In an embodiment, one reception circuit 323 may be configured to be electrically connected to two or more VCOs. In an embodiment, as the sizes of the elements included in the RFIC 320 decrease, the distance between the VCOs may decrease. When the VCOs are disposed close to each other, coupling may occur due to the inductor included in the VCO. For example, coupling 421 may occur between the transmission VCO 401 and the reception VCO 403. In an embodiment, the electronic device 101 (e.g., the communication processor 310) may identify whether the signal output from the reception VCO 403 is an abnormal signal, based on the operation of the signal detection block 411. The electronic device 101 may control the RFIC 320 to reduce coupling between the VCOs based on identifying that the signal output from the received VCO 403 is an abnormal signal. For example, the electronic device 101

may control the RFIC 320 to change the operating frequency of the transmission VCO 401. An operation in which the electronic device 101 controls the RFIC 320 is described below with reference to FIGS. 5 to 11. In an embodiment, the electronic device 101 may control RF circuit to turn on the signal detection block 411, based on identifying that the transmission circuit 321 operates. Based on identifying that the transmission circuit 321 operates stably, the electronic device 101 may control the RF circuit to turn off the signal detection block 411. For example, the electronic device 101 may control the signal detection block 411 to be turned off after the frequency of the transmission circuit 321 is locked. Based on identifying that the frequency of the transmission circuit 321 is locked, the electronic device 101 may control the signal detection block 411 to be turned off. The electronic device 101 may reduce unnecessary current consumption of the signal detection block 411 based on controlling the signal detection block 411 to be turned on or off.

[0070] In an embodiment of the disclosure, the transmission VCO 401 and/or the reception VCO 403 may be referred to as a “first VCO” and/or a “second VCO”. In the disclosure, the first VCO or the second VCO is not limited to the transmission VCO 401 or the reception VCO 403. In an embodiment, the electronic device 101 may control the RFIC 320 such that the coupling 421 between the transmission VCO 401 and the reception VCO 403 is reduced. In an embodiment, the electronic device 101 may control at least one RFIC (e.g., at least one of the first RFIC 222, the second RFIC 224, the third RFIC 226, the fourth RFIC 228 of FIGS. 2A and 2B, or the RFIC 320 of FIG. 4) of the electronic device 101 to reduce coupling between different transmission VCOs. The electronic device 101 may control the RFIC 320 to reduce coupling between different reception VCOs.

[0071] FIG. 5 is a view illustrating each component included in an RFIC according to an embodiment of the disclosure.

[0072] In an embodiment, the transmission circuit 321 included in the RFIC 320 (e.g., at least one of the first RFIC 222, the second RFIC 224, the third RFIC 226, the fourth RFIC 228 of FIG. 2A, the RFIC 320 of FIG. 3, or the RFIC 320 of FIG. 4) may be configured to include digital-to-analog converters (DACs) 511a and 511b, low-pass filters (hereinafter referred to as “LPFs”) 512a and 512b, voltage-to-current (V2I) 513a and 513b, mixers 514a and 514b, or an amplifier 515. In an embodiment, the DACs 511a and 511b may convert a digital signal received from the communication processor 310 into an analog signal. The LPFs 512a and 512b may attenuate noise of the analog signal output from the DACs 511a and 511b according to the bandwidth of the transmission signal in the baseband band. The V2Is 513a and 513b may convert a voltage signal into a current signal and transmit the current signal to the mixers 514a and 514b. The mixers 514a and 514b may convert a baseband signal into an RF signal. The amplifier 515 may adjust the power of the transmission RF signal. The amplifier 515 may be implemented as a voltage gain amplifier (VGA) and/or a drive amplifier (DA), but is not limited thereto.

[0073] In an embodiment, the reception circuit 323 included in the RFIC 320 may be configured to include an LNA, mixers 524a and 524b, transimpedance amplifiers (TIAs) 523a and 523b, LPFs 522a and 522b, and analog-to-digital converters (ADCs) 521a and 521b. The LNA may amplify an RF signal received through an antenna module (e.g., the antenna module 340 of FIG. 3). The mixers 524a and 524b may convert an RF signal into a baseband signal. The TIAs 523a and 523b may convert a current signal output from the mixers 524a and 524b into a voltage signal. The LPFs 522a and 522b may attenuate noise of an ambient band of a received signal in a baseband. The ADCs 521a and 521b may convert an analog signal output from the LPFs 522a and 522b into a digital signal.

[0074] In an embodiment, the reception VCO 403 included in the RFIC 320 may be configured to include a LOIQ 551 and a phase-locked loop (PLL) 560. The LOIQ 551 may be configured to include a signal generator and a frequency divider for generating an IQ signal. For example, the frequency divider may include a $\frac{1}{2}$ frequency divider and/or a $\frac{1}{4}$ frequency divider. The PLL 560

may include a buffer **564**, a phase comparator **562**, an LPF **561**, and a 1/N frequency divider **563**. The buffer **564** may increase the signal strength of the reference clock Ref CLK. The phase comparator **562** may adjust the frequency based on frequency comparison. The LPF **561** may reduce phase noise based on filtering the signal. The 1/N frequency divider **563** may identify the signal output from the reception VCO **403**.

[0075] In an embodiment, the transmission VCO **401** included in the RFIC **320** may be configured to include a LOIQ **531** and a PLL **540**. The LOIQ **531** may be configured to include a signal generator and a frequency divider for generating an IQ signal. For example, the frequency divider may include a $\frac{1}{2}$ frequency divider and/or a $\frac{1}{4}$ frequency divider. The PLL **540** may include a buffer **525**, at least one frequency divider **544a** and **544b**, a phase comparator **542**, an LPF **541**, and a 1/N frequency divider **543**. The buffer **525** may increase the signal strength of the reference clock. The at least one frequency divider **544a** and **544b** may change the reference clock. The at least one frequency divider **544a** and **544b** may include a $\frac{1}{2}$ frequency divider **544a** and/or a $\frac{2}{3}$ frequency divider **544b**. In an embodiment, the $\frac{1}{2}$ frequency divider **544a** and the $\frac{2}{3}$ frequency divider **544b** may be connected in parallel. In an embodiment, the bypass path **545** may be connected in parallel with the $\frac{1}{2}$ frequency divider **544a** and the $\frac{2}{3}$ frequency divider **544b**. When frequency division is not required, the electronic device **101** may control the RFIC **320** so that the signal is transmitted through the bypass path **545**. For example, when there is no signal interference due to VCO coupling, the electronic device **101** may control the RFIC **320** so that the signal is transmitted to the bypass path **545**. In an embodiment, a state in which a signal is transmitted through the bypass path **545** may be referred to as a “bypass mode”. The phase comparator **542** may adjust the frequency based on the frequency comparison. The LPF **541** may reduce phase noise based on filtering the signal. The 1/N frequency divider **543** may identify the signal output from the transmission VCO **401**. In an embodiment, the transmission VCO **401** may be configured to operate at a frequency twice as high as the target frequency. For example, the transmission VCO **401** may output a signal of 4.2 GHz so that the electronic device **101** transmits a signal of 2.1 GHz in the B1 band.

[0076] In an embodiment, the signal detection block **411** included in the RFIC **320** may be configured to include a 1/N frequency divider **573**, an LPF **572**, or an ADC **571**. The 1/N frequency divider **573** may convert an RF signal into a baseband signal. The LPF **572** may output only a signal of a band corresponding to the reception circuit **323**. LPF **572** may be configured to further include a VGA, but is not limited thereto. The VGA may adjust the power of the signal output from the LPF **572**. The ADC **571** may convert an analog signal output from the LPF **572** into a digital signal. The electronic device **101** (e.g., the communication processor **310**) may identify the quality of the LO signal output from the LOIQ **551** of the reception VCO **403** and input to the reception circuit **323** based on the operation of the signal detection block **411**.

[0077] According to an aspect of the disclosure, an electronic device is presented which comprises a memory storing instructions, a radio-frequency (RF) circuit, and a communication processor operatively connected to the memory and the RF circuit. The RF circuit may include a first voltage-controlled oscillator (VCO), a second VCO, and a signal detection block electrically connected to the first VCO and configured to detect (or identify) a signal output from the first VCO. The RF circuit may preferably further comprise a reception circuit (e.g. an Rx chain) and/or a transmission circuit (e.g. a Tx chain). The first VCO may be a reception VCO electrically connected to a reception circuit of the RF circuit. The second VCO may be a transmission VCO electrically connected to a transmission circuit of the RF circuit.

[0078] The instructions stored by the memory may preferably cause the electronic device, when executed by the communication processor, to control the RF circuit to change an operating frequency of the second VCO based on identifying that a signal output from the first VCO is an abnormal signal (i.e. is abnormal) when the strength of the signal output from the first VCO as detected by the signal detection block is less than a reference value.

[0079] The first VCO may comprise an LOIQ (i.e. a local oscillator in-phase/quadrature generator). The signal detection block may be electrically connected to the LOIQ of the first VCO.

Alternatively or additionally, the second VCO may comprise an LOIQ.

[0080] The LOIQ may comprise or consist of a generator to form IQ signals and optionally a frequency divider. The transmission and/or reception circuits may be configured to analyze or process IQ signals or I/Q data, i.e. in-phase and quadrature components of an electrical signal.

[0081] The signal detection block may be implemented as a local oscillator (LO) signal quality detection block. For example, the role of the LO signal quality detection block can be to check the quality of the LO signal passing from the first VCO (e.g. through the LOIQ thereof) to the reception circuit.

[0082] The signal detection block may include a frequency divider electrically connected to the first VCO, a low pass filter electrically connected to the frequency divider, and an analog-to-digital converter (ADC) electrically connected to the low pass filter. Preferably, a variable gain amplifier may be electrically connected to the low pass filter. In this case, the ADC may be electrically connected to the variable gain amplifier. The variable gain amplifier can regulate any lack of power so that the signal can correctly reach the ADC. The ADC may be at a terminal position, i.e. may be a final ADC.

[0083] Preferably, the electronic device may further comprise a third VCO. The signal detection block may then include a mixer that is electrically connected to the first VCO (e.g. the LOIQ thereof) and to the third VCO, preferably an LOIQ of the third VCO. A low pass filter may be electrically connected to the mixer and an ADC may be electrically connected to the low pass filter. Optionally, a variable gain amplifier may be electrically connected to the low pass filter and the ADC may be electrically connected to the variable gain amplifier.

[0084] Preferably, the signal detection block may be electrically connected to the LOIQ of the first VCO. In this case, it may be further preferred that the instructions, when executed by the communication processor, cause the electronic device to detect a signal output from the LOIQ of the first VCO with the signal detection block, identify whether the detected signal output from the LOIQ of the first VCO is an abnormal signal based on identifying a strength of the detected signal (e.g. the signal output from the LOIQ of the first VCO as detected by the signal detection block is less than a reference value), and control the RF circuit to change the operating frequency of the second VCO when identifying that the detected signal is abnormal.

[0085] Preferably, the instructions, when executed by the communication processor, may cause the electronic device to identify the strength of the detected signal based on identifying a signal-to-interference plus-noise ratio (SINR) of the detected signal. The SINR is an example to identify the strength of a signal and other metrics could be used as alternative or in addition.

[0086] Preferably, the second VCO may include a phase-locked loop (PLL). The PLL may include a buffer receiving a reference clock, at least one frequency divider electrically connected to the buffer, a phase comparator electrically connected to the at least one frequency divider, and a low pass filter electrically connected to the phase comparator. The at least one frequency divider of the PLL may be implemented as one or more frequency dividers connected in parallel between the buffer and the phase comparator.

[0087] When the RF circuit comprises the or at least one reception circuit, it may be preferred that the instructions, when executed by the communication processor, cause the electronic device to control at least one frequency divider of the PLL of the second VCO to change a reference clock of the second VCO based on identifying that the detected signal is abnormal, and/or to control the RF circuit to change an operating frequency of the second VCO when identifying that a strength of a signal received through the reception circuit is less than a reference value.

[0088] Alternatively or additionally yet also when the RF circuit comprises the reception circuit, the instructions, when executed by the communication processor, may be configured to cause the electronic device to control the RF circuit to change the operating frequency of the second VCO

based on identifying that the SINR of the signal received through the reception circuit is less than a reference value.

[0089] Further alternatively or additionally when the RF circuit comprises the reception circuit, the instructions, when executed by the communication processor, may be configured to cause the electronic device to control the RF circuit to change the operating frequency of the second VCO based on identifying that a reference signal received power (RSRP) of the signal received through the reception circuit is a reference value or more.

[0090] In particular, it may be envisaged to identify both the SINR as well as the RSRP of the signal received through the reception circuit and control the RF circuit to change the operating frequency of the second VCO based on identifying that the RSRP of the signal received through the reception circuit is a reference value or more and that the SINR of the signal is less than a reference value. In other words, the operating frequency of the second VCO (e.g. reception VCO) will be changed if the SINR is low despite a high RSRP.

[0091] In an embodiment, the instructions, when executed by the communication processor, may cause the electronic device to identify a lookup table associated with an operating frequency combination of the electronic device based on identifying that the detected signal is abnormal, control the RF circuit to change the operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination, and control at least one frequency divider of the PLL of the second VCO to change a reference clock of the second VCO based on identifying that the operating frequency combination is a second frequency combination and/or is not the first frequency combination.

[0092] When the RF circuit comprises the or at least one transmission circuit, it may be preferred that the instructions, when executed by the communication processor, cause the electronic device to control the RF circuit to turn on the signal detection block based on identifying that the transmission circuit operates and/or to turn off the signal detection block based on identifying that a frequency of the transmission circuit is locked.

[0093] FIGS. 6A, 6B, and 6C are views illustrating an example of coupling between VCOs according to an embodiment of the disclosure.

[0094] Referring to FIG. 6A, a frequency band in which coupling between a B3 transmission VCO and a B3 reception VCO occurs is illustrated. Referring to FIG. 6A, the B3 transmission VCO may output a signal **601** of 3484 MHz for a target frequency of 1742 MHz. In an embodiment, the reference clock of the B3 transmission VCO may have 36 MHz (Δf). In an embodiment, due to the operation of the 1/N frequency divider inside the PLL, a spurious (spur) signal, which is a signal of an unintended frequency band, may be generated in the VCO output. The spur signal may mean an unwanted signal generated in a frequency band other than the frequency to be output. For example, the five-fold frequency of the reference clock and the operating frequency of the B3 transmission VCO may form a spur signal **603** of 3664 MHz. In an embodiment, the B3 reception VCO may output a signal **611** of 3674 MHz for a target frequency of 1837 MHz. Since the output signal of the B3 reception VCO and the spur signal of the B3 transmission VCO have a frequency difference (Δf_1) of about 20 MHz or less of bandwidth, coupling between the B3 transmission VCO and the B3 reception VCO may occur.

[0095] Referring to FIG. 6B, a frequency band in which coupling between a B28 reception VCO and a B20 reception VCO occurs is illustrated. Referring to FIG. 6B, the B28 reception VCO may output a signal **621** of 3100 MHz for a target frequency of 775 MHz. In an embodiment, the reference clock of the B28 reception VCO may have 36 MHz (Δf). The three-fold frequency of the reference clock and the operating frequency of the B20 reception VCO may form a spur signal **623** of 3208 MHz. In an embodiment, the B20 reception VCO may output a signal **631** of 3204 MHz, which is four times the target frequency, for a target frequency of 801 MHz. Since the output signal of the B20 reception VCO and the spur signal of the B28 reception VCO have a frequency difference (Δf_2) of about 20 MHz or less of bandwidth, coupling between the B28 reception VCO

and the B20 reception VCO may occur.

[0096] Referring to FIG. 6C, a frequency band in which coupling between a B3 reception VCO and an N78 reception VCO occurs in an EN-DC combination due to LO leakage is illustrated.

Referring to FIG. 6C, the B3 reception VCO may output a signal **641** of 3674 MHz for a target frequency of 1837 MHz. In an embodiment, the N78 reception VCO may output a signal **651** of 7260 MHz, which is twice the target frequency, for a signal **653** of a target frequency of 3630 MHz. Since the frequency of the output signal of the B3 reception VCO and the target frequency of the N78 reception VCO have a frequency difference (Δf_3) of about 100 MHz or less of bandwidth, coupling between the B3 reception VCO and the N78 reception VCO may occur.

[0097] FIG. 7 is a block diagram illustrating a signal detection block according to an embodiment of the disclosure.

[0098] In an embodiment, the signal detection block **411** may be configured to include a mixer **713**, an LPF **712**, or an ADC **711**. When the signal detection block **411** is configured to include a mixer **713** rather than the 1/N frequency divider (e.g., the frequency divider **573** of FIG. 5) of the signal detection block **411** of FIG. 5, the RFIC (e.g., the RFIC **320** of FIG. 3) may be configured to further include an additional VCO **720** electrically connected to the mixer **713**. The additional VCO **720** may be configured to include an LOIQ **721** and a PLL **730**. The signal detection block **411** including the mixer **713** may more accurately detect an abnormal signal as the band of the LPF **712** becomes substantially the same as the band of the reception circuit (e.g., the reception circuit **323** of FIG. 3) based on converting the reception RF signal with respect to 0 Hz.

[0099] FIG. 8 is a flowchart **800** illustrating an operation of changing an operating frequency of a VCO of an electronic device according to an embodiment of the disclosure.

[0100] Referring to FIG. 8, in operation **801**, in an embodiment, the electronic device (the electronic device **101** of FIG. 1) (e.g., at least one of the processor **120** of FIG. 1, the first communication processor **212** of FIG. 2A, the second communication processor **214**, the integrated communication processor **260** of FIG. 2B, or the communication processor **310** of FIG. 3) may identify a signal output from the first VCO. For example, The electronic device **101** may identify a signal output from the LOIQ (e.g., the LOIQ **551** of FIG. 5) of the reception VCO (e.g., the reception VCO **403** of FIG. 4) based on the signal detection block (e.g., the signal detection block **411** of FIG. 4).

[0101] In operation **803**, in an embodiment, the electronic device **101** may identify whether the identified signal is an abnormal signal. In an embodiment, the electronic device **101** may identify the strength of the identified signal based on identifying the signal-to-interference plus-noise ratio (SINR) output from the LOIQ **551**. When the SINR is smaller than a reference value, the electronic device **101** may identify that it is an abnormal signal. The reference value for identifying that it is an abnormal signal may be changed according to an embodiment, and is not limited to a specific value.

[0102] In operation **805**, in an embodiment, the electronic device **101** may change the operating frequency of the second VCO. For example, the electronic device **101** may control to change the operating frequency of the transmission VCO (e.g., the transmission VCO **401** of FIG. 4) based on identifying that the signal output from the LOIQ **551** is an abnormal signal. The electronic device **101** may control to change the operating frequency of the transmission VCO **401** based on selecting an operating frequency different from the frequency band in which the coupling occurs.

[0103] FIG. 9 is a flowchart **900** illustrating an operation of changing an operating frequency of a VCO based on a strength of a signal received through a reception circuit by an electronic device according to an embodiment of the disclosure.

[0104] Referring to FIG. 9, in operation **901**, in an embodiment, the electronic device (the electronic device **101** of FIG. 1) (e.g., at least one of the processor **120** of FIG. 1, the first communication processor **212** of FIG. 2A, the second communication processor **214**, the integrated communication processor **260** of FIG. 2B, or the communication processor **310** of FIG. 3) may

identify a signal output from the first VCO. For example, the electronic device **101** may identify a signal output from the LOIQ (e.g., the LOIQ **551** of FIG. 5) of the reception VCO (e.g., the reception VCO **403** of FIG. 4) based on the signal detection block (e.g., the signal detection block **411** of FIG. 4). Since operation **901** is at least partially the same as or similar to operation **801**, the same description as that of FIG. 8 may not be repeated.

[0105] In operation **903**, in an embodiment, the electronic device **101** may identify whether the identified signal is an abnormal signal. Since operation **903** is at least partially the same as or similar to operation **803**, the same description as that of FIG. 8 may not be repeated.

[0106] In operation **905**, in an embodiment, the electronic device **101** may change the reference clock. In an embodiment, the electronic device **101** may change the reference clock based on the frequency divider (e.g., $\frac{1}{2}$ frequency divider **544a** and/or $\frac{2}{3}$ frequency divider **544b** of FIG. 5) so that the operating frequency of the transmission VCO (e.g., the transmission VCO **401** of FIG. 4) is adjacent to an even number-fold frequency of the operating frequency of the reception VCO **403**.

[0107] In operation **907**, in an embodiment, the electronic device **101** may identify whether the strength of the signal received through the reception circuit (e.g., the reception circuit **323** of FIG. 4) is less than the reference value. For example, the electronic device **101** may identify whether the SINR of the signal received through the reception circuit **323** is less than the reference value.

[0108] In an embodiment, the electronic device **101** may identify the signal-to-interference plus-noise ratio (SINR) and the reference signal received power (RSRP) of the signal received through the reception circuit **323**. The electronic device **101** may identify whether the RSRP of the signal is greater than or equal to a reference value, and whether the SINR of the signal is less than a reference value. The electronic device **101** may identify whether the transmission channel and/or the reception channel of the RF signal are good based on identifying the RSRP value. For example, the electronic device **101** may identify that the transmission channel and/or the reception channel of the RF signal are good based on identifying that the RSRP value is equal to or greater than the reference value. The electronic device **101** may identify that VCO coupling occurs based on identifying that the RSRP of the signal received through the reception circuit **323** is equal to or greater than the reference value and the SINR of the reception signal is less than the reference value.

[0109] In operation **909**, in an embodiment, the electronic device **101** may change the operating frequency of the second VCO. For example, the electronic device **101** may change the operating frequency of the transmission VCO **401** based on the condition of operation **907** being met. Since operation **909** is at least partially the same as or similar to operation **805**, the same description as that of FIG. 8 may not be repeated.

[0110] FIG. 10 is a flowchart **1000** illustrating an operation of changing an operating frequency of a VCO based on identifying a lookup table by an electronic device according to an embodiment of the disclosure.

[0111] Referring to FIG. 10, in operation **1001**, in an embodiment, the electronic device (**101** of FIG. 1) (e.g., at least one of the processor **120** of FIG. 1, the first communication processor **212** of FIG. 2A, the second communication processor **214**, the integrated communication processor **260** of FIG. 2B, or the communication processor **310** of FIG. 3) may identify a signal output from the first VCO. Since operation **1001** is at least partially the same as or similar to operation **801**, the same description as that of FIG. 8 may not be repeated.

[0112] In operation **1003**, in an embodiment, the electronic device **101** may identify whether the identified signal is an abnormal signal. Since operation **1003** is at least partially the same as or similar to operation **803**, the same description as that of FIG. 8 may not be repeated.

[0113] In operation **1005**, in an embodiment, the electronic device **101** may identify the lookup table. In an embodiment, the electronic device **101** may identify a lookup table (LUT) associated with an operating frequency combination of the electronic device based on identifying that the identified signal is an abnormal signal. For example, the lookup table may be stored in the

electronic device **101**.

[0114] In operation **1007**, in an embodiment, the electronic device **101** may change the operating frequency of the second VCO. The electronic device **101** may control at least one RFIC (e.g., at least one of the first RFIC **222**, the second RFIC **224**, the third RFIC **226**, the fourth RFIC **228** of FIG. 2A, or the RFIC **320** of FIG. 3) to change the operating frequency of the transmission VCO (e.g., the transmission VCO **401** of FIG. 4) based on identifying that the operating frequency combination is a first frequency combination stored in the lookup table.

[0115] FIG. 11 is a flowchart **1100** illustrating an operation of changing a reference clock based on identifying a lookup table by an electronic device according to an embodiment of the disclosure.

[0116] Referring to FIG. 11, in operation **1101**, in an embodiment, the electronic device (**101** of FIG. 1) (e.g., at least one of the processor **120** of FIG. 1, the first communication processor **212** of FIG. 2A, the second communication processor **214**, the integrated communication processor **260** of FIG. 2B, or the communication processor **310** of FIG. 3) may identify a signal output from the first VCO. Since operation **1101** is at least partially the same as or similar to operation **801**, the same description as that of FIG. 8 may not be repeated.

[0117] In operation **1103**, in an embodiment, the electronic device **101** may identify whether the identified signal is an abnormal signal. Since operation **1103** is at least partially the same as or similar to operation **803**, the same description as that of FIG. 8 may not be repeated.

[0118] In operation **1105**, in an embodiment, the electronic device **101** may identify the lookup table. Since operation **1105** is at least partially the same as or similar to operation **1005**, the same description as that of FIG. 10 may not be repeated.

[0119] In operation **1107**, in an embodiment, the electronic device **101** may change the reference clock. The electronic device **101** may control at least one frequency divider (e.g., the $\frac{1}{2}$ frequency divider **544a** and/or $\frac{2}{3}$ frequency divider **544b** of FIG. 5) of the PLL (e.g., the PLL **540** of FIG. 5) to change the reference clock of the transmission VCO (e.g., the transmission VCO **401** of FIG. 4) based on identifying that the operating frequency combination is a second frequency combination.

[0120] The electronic device **101** may control at least one frequency divider **544a** and **544b** of the PLL **540** to change the reference clock of the transmission VCO **401** for the remaining combinations, based on identifying that the operation frequency combination is not the first frequency combination.

[0121] According to a further aspect of the disclosure, a method for operating an electronic device is presented, the method may comprise operations of detecting (or identifying) a signal output from a first VCO included in an RF circuit of the electronic device through a signal detection block included in the RF circuit, identifying whether the detected signal is an abnormal signal based on identifying a strength of the detected signal, and controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the identified signal is abnormal when the strength of the detected signal is less than a reference value.

[0122] Preferably, the method may further comprise controlling at least one frequency divider included in a PLL of the second VCO to change a reference clock of the second VCO based on identifying that the identified signal is abnormal, and controlling the RF circuit to change an operating frequency of the second VCO based on identifying that a strength of a signal received through a reception circuit included in the RF circuit is less than a reference value.

[0123] Preferably, the method may further comprise identifying a lookup table associated with an operating frequency combination of the electronic device based on identifying that the identified signal is abnormal, controlling the RF circuit to change the operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination, and controlling at least one frequency divider included in a PLL of the second VCO to change a reference clock of the second VCO based on identifying that the operating frequency combination is a second frequency combination and/or identifying that the operating frequency combination is not the first frequency combination.

[0124] Preferably, the method may further comprise controlling the RF circuit to turn on the signal detection block based on identifying that a transmission circuit included in the RF circuit operates and/or controlling the RF circuit to turn off the signal detection block based on identifying that a frequency of a transmission circuit included in the RF circuit is locked. In this way, the signal detection block or LO signal quality detection block can be activated and/or deactivated during certain protocol scenarios such as a locking operation to minimize current.

[0125] The electronic device as disclosed herein can be configured to perform any of the operations of the method according to the present disclosure. Likewise, the method may involve any of the functions disclosed for (components of) the electronic device. Further, operations and/or functions described in relation to the method or electronic device can equally apply for the instructions stored by the non-transitory storage medium.

[0126] The electronic device and method serve to reduce VCO coupling effects and can in particular prevent spur and/or LO leakage from crossing to a reception circuit or Rx chain coupled or couplable to the first VCO. Sensitivity degradation and throughput reduction can be prevented. When VCO coupling effects are reduced, VCOs can be positioned closer together, thus enabling more efficient use of space in an electronic device and enable further miniaturization.

[0127] According to an embodiment of the disclosure, an electronic device (e.g., the electronic device **101** of FIG. **1**) may be configured to include at least one RFFE (e.g., the first RFFE **232**, the second RFFE **234**, the third RFFE **236**, and the RFFE **330** of FIG. **2A** and FIG. **2B**), an RF circuit electrically connected to the at least one RFFE **232**, **234**, **236**, and **330**, and a communication processor (e.g., the processor **120** of FIG. **1**, the first communication processor **212** or the second communication processor **214** of FIG. **2A**, the integrated communication processor **260** of FIG. **2B**, or the communication processor **310** of FIG. **3**) operatively connected to the at least one RFFE **232**, **234**, **236**, and **330** and the RF circuit. The RF circuit of the electronic device **101** may be configured to include at least one reception circuit (e.g., the reception circuit **323** of FIG. **4**), at least one transmission circuit (e.g., the transmission circuit **321** of FIG. **4**), a first VCO, a second VCO, and a signal detection block (e.g., the signal detection block **411** of FIG. **4**) electrically connected to the first VCO. The communication processor **120**, **212**, **214**, **260**, or **310** of the electronic device **101** may be configured to control the RF circuit to change an operating frequency of the second VCO based on identifying whether a signal output from the first VCO is an abnormal signal.

[0128] In an embodiment, the signal detection block **411** of the electronic device **101** may be configured to include a frequency divider (e.g., the 1/N frequency divider **573** of FIG. **5**) electrically connected to an LOIQ (e.g., the LOIQ **551** of FIG. **5**) of the first VCO, a low pass filter electrically connected to the frequency divider **573**, a variable gain amplifier electrically connected to the low pass filter, and an analog-to-digital (ADC) converter (e.g., the ADC **571** of FIG. **5**) electrically connected to the variable gain amplifier.

[0129] In an embodiment, the electronic device **101** may be configured to further include a third VCO (e.g., the additional VCO **720** of FIG. **7**). The signal detection block **411** of the electronic device **101** may be configured to include a mixer (e.g., the mixer **713** of FIG. **7**) electrically connected to the LOIQ **551** of the first VCO and an LOIQ (e.g., the LOIQ **721** of FIG. **7**) of the third VCO **720**, a low pass filter electrically connected to the mixer **713**, a variable gain amplifier electrically connected to the low pass filter, and an ADC (e.g., the ADC **711** of FIG. **7**) electrically connected to the variable gain amplifier.

[0130] In an embodiment, the communication processor **120**, **212**, **214**, **260**, or **310** of the electronic device **101** may be configured to identify a signal output from an LOIQ of the first VCO. In an embodiment, the communication processor **120**, **212**, **214**, **260**, or **310** of the electronic device **101** may be configured to identify whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. In an embodiment, the communication processor **120**, **212**, **214**, **260**, or **310** of the electronic device **101** may be configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that the identified signal

is the abnormal signal.

[0131] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be configured to identify the strength of the identified signal based on identifying an SINR of the identified signal.

[0132] In an embodiment, the second VCO of the electronic device **101** may be configured to include a phase-locked loop (PLL) (e.g., the PLL **540** of FIG. 5). The PLL **540** may be configured to include a buffer (e.g., the buffer **525** of FIG. 5) receiving a reference clock, at least one frequency divider (e.g., the $\frac{1}{2}$ frequency divider **544a** or the $\frac{2}{3}$ frequency divider **544b** of FIG. 5) electrically connected to the buffer **525**, a phase comparator (e.g., the phase comparator **542** of FIG. 5) electrically connected to the at least one frequency divider **544a** and **544b**, and a low pass filter (e.g., the low pass filter **541** of FIG. 5) electrically connected to the phase comparator **542**.

[0133] In an embodiment, the at least one frequency divider **544a** and **544b** of the PLL **540** of the electronic device **101** may be one or more frequency dividers **544a** and **544b** connected in parallel between the buffer **525** and the phase comparator **542**.

[0134] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify a signal output from an LOIQ of the first VCO. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control at least one frequency divider **544a** and **544b** of the PLL **540** to change a reference clock of the second VCO based on identifying that the identified signal is an abnormal signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that the strength of the signal received through the reception circuit **403** is less than a reference value.

[0135] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that an SINR of the signal received through the reception circuit **403** is less than a reference value.

[0136] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be configured to identify the SINR and RSRP of the signal received through the reception circuit **403**. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that the RSRP of the signal is a reference value or more and/or the SINR of the signal is less than a reference value.

[0137] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify a signal output from an LOIQ of the first VCO. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify a lookup table associated with an operating frequency combination of the electronic device **101** based on identifying that the identified signal is the abnormal signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control at least one frequency divider **544a** and **544b** of the PLL **540** to change a reference clock of the second VCO based on identifying that the

operating frequency combination is a second frequency combination.

[0138] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify a signal output from an LOIQ of the first VCO. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to identify a lookup table associated with an operating frequency combination of the electronic device based on identifying that the identified signal is the abnormal signal. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to change an operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination. In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control at least one frequency divider **544a** and **544b** of the PLL **540** to change a reference clock of the second VCO based on identifying that the operating frequency combination is not the first frequency combination.

[0139] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to turn on the signal detection block **411** based on identifying that the transmission circuit **321** operates.

[0140] In an embodiment, the communication processor **120, 212, 214, 260, or 310** of the electronic device **101** may be further configured to control the RF circuit to turn off the signal detection block **411** based on identifying that the frequency of the transmission circuit **321** is locked.

[0141] According to an embodiment of the disclosure, a method for operating an electronic device **101** may include identifying a signal output from a first VCO included in an RF circuit of the electronic device **101** through a signal detection block **411** included in the RF circuit. The method for operating the electronic device **101** may include identifying whether the identified signal is an abnormal signal based on identifying a strength of the identified signal. The method for operating the electronic device **101** may include controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the identified signal is an abnormal signal.

[0142] In an embodiment, the method for operating the electronic device **101** may further include controlling at least one frequency divider **544a** and **544b** included in a PLL **540** of the second VCO to change a reference clock of the second VCO based on identifying that the identified signal is the abnormal signal. In an embodiment, the method for operating the electronic device **101** may further include controlling the RF circuit to change an operating frequency of the second VCO based on identifying that a strength of a signal received through a reception circuit **323** included in the RF circuit is less than a reference value.

[0143] In an embodiment, the method for operating the electronic device **101** may further include identifying a lookup table associated with an operating frequency combination of the electronic device **101** based on identifying that the identified signal is the abnormal signal. In an embodiment, the operation method may further include controlling the RF circuit to change an operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination. The method for operating the electronic device **101** may further include controlling at least one frequency divider **544a** and **544b** included in a PLL **540** of the second VCO to change a reference clock of the second VCO based on identifying that the operating frequency combination is a second frequency combination.

[0144] In an embodiment, the method for operating the electronic device **101** may further include identifying a lookup table associated with an operating frequency combination of the electronic device **101** based on identifying that the identified signal is the abnormal signal. In an embodiment,

the operation method may further include controlling the RF circuit to change an operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination. The method for operating the electronic device **101** may further include controlling at least one frequency divider **544a** and **544b** included in a PLL **540** of the second VCO to change a reference clock of the second VCO based on identifying that the operating frequency combination is not the first frequency combination.

[0145] In an embodiment, the method for operating the electronic device **101** may further include controlling the RF circuit to turn on the signal detection block **411** based on identifying that the transmission circuit **321** included in the RF circuit operates.

[0146] In an embodiment, the method for operating the electronic device **101** may further include controlling the RF circuit to turn off the signal detection block **411** based on identifying that the frequency of the transmission circuit **321** included in the RF circuit is locked.

[0147] The electronic device according to an embodiment of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0148] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0149] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0150] An embodiment of the disclosure may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The storage medium readable by the machine may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage

medium and where the data is temporarily stored in the storage medium.

[0151] According to an embodiment, a method according to an embodiment of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smartphones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0152] According to an embodiment, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to an embodiment, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or Further, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0153] Further, the structure of the data used in embodiments of the disclosure may be recorded in a computer-readable recording medium via various means. The computer-readable recording medium includes a storage medium, such as a magnetic storage medium (e.g., a ROM, a floppy disc, or a hard disc) or an optical reading medium (e.g., a CD-ROM or a DVD).

[0154] Example embodiments of the disclosure have been described above. The above-described embodiments are merely examples, and it will be appreciated by one of ordinary skill in the art various changes may be made thereto without departing from the scope of the present invention. Therefore, the disclosed embodiments should be considered from an illustrative, rather than a limiting, point of view. The scope of the invention is indicated in the claims rather than in the above-described description. All differences within the equivalent range should be construed as being included in the present invention.

Claims

1. An electronic device comprising: memory storing instructions; a radio-frequency, RF, circuit; and a communication processor operatively connected to the memory and the RF circuit, wherein the RF circuit includes: a first voltage-controlled oscillator, VCO; a second voltage-controlled oscillator, VCO; and a signal detection block electrically connected to the first VCO and configured to detect a signal output from the first VCO, and wherein the instructions, when executed by the communication processor, cause the electronic device to control the RF circuit to change an operating frequency of the second VCO based on identifying that a signal output from the first VCO is an abnormal signal when the strength of the signal output from the first VCO as detected by the signal detection block is less than a reference value.
2. The electronic device of claim 1, wherein the signal detection block includes: a frequency divider electrically connected to the first VCO, preferably an LOIQ thereof; a low pass filter electrically connected to the frequency divider; optionally, a variable gain amplifier electrically connected to the low pass filter; and an analog-to-digital converter, ADC, electrically connected to the low pass filter or to the variable gain amplifier when present.

3. The electronic device of claim 1, further comprising a third VCO, wherein the signal detection block includes: a mixer electrically connected to the first VCO, preferably the LOIQ thereof, and the third VCO, preferably an LOIQ thereof; a low pass filter electrically connected to the mixer; optionally, a variable gain amplifier electrically connected to the low pass filter; and an ADC electrically connected to the low pass filter or to the variable gain amplifier when present.
4. The electronic device of claim 1, wherein the signal detection block is electrically connected to an LOIQ of the first VCO and the instructions, when executed by the communication processor, cause the electronic device to: detect a signal output from the LOIQ of the first VCO with the signal detection block; identify whether the detected signal output from the LOIQ of the first VCO is an abnormal signal based on identifying a strength of the detected signal; and control the RF circuit to change the operating frequency of the second VCO when identifying that the detected signal is abnormal.
5. The electronic device of claim 1, wherein the instructions, when executed by the communication processor, cause the electronic device to: identify the strength of the detected signal based on identifying a signal-to-interference plus-noise ratio, SINR, of the detected signal.
6. The electronic device of claim 1, wherein the second VCO includes a phase-locked loop, PLL, and wherein the PLL includes: a buffer receiving a reference clock; at least one frequency divider electrically connected to the buffer; a phase comparator electrically connected to the at least one frequency divider; and a low pass filter electrically connected to the phase comparator, wherein optionally the at least one frequency divider of the PLL is one or more frequency dividers connected in parallel between the buffer and the phase comparator.
7. The electronic device of claim 1, wherein the RF circuit further comprises a reception circuit and the instructions, when executed by the communication processor, cause the electronic device to: control at least one frequency divider of a PLL of the second VCO to change a reference clock of the second VCO based on identifying that the detected signal is abnormal; and control the RF circuit to change an operating frequency of the second VCO when identifying that a strength of a signal received through the reception circuit is less than a reference value.
8. The electronic device of claim 1, wherein the RF circuit further comprises a reception circuit and the instructions, when executed by the communication processor, cause the electronic device to: control the RF circuit to change the operating frequency of the second VCO based on identifying that a signal-to-interference plus-noise ratio, SINR, of the signal received through the reception circuit is less than a reference value.
9. The electronic device of claim 1, wherein the RF circuit further comprises a reception circuit and the instructions, when executed by the communication processor, cause the electronic device to control the RF circuit to: change the operating frequency of the second VCO based on identifying that a reference signal received power, RSRP, of the signal received through the reception circuit is a reference value or more.
10. The electronic device of claim 1, wherein the instructions, when executed by the communication processor, cause the electronic device to: identify a lookup table associated with an operating frequency combination of the electronic device based on identifying that the detected signal is abnormal; control the RF circuit to change the operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination; and control at least one frequency divider of a PLL of the second VCO to change a reference clock of the second VCO based on identifying that at least one of: the operating frequency combination is a second frequency combination; or the operating frequency combination is not the first frequency combination.
11. The electronic device of claim 1, further comprising at least one transmission circuit and wherein the instructions, when executed by the communication processor, cause the electronic device to control the RF circuit to at least one of: turn on the signal detection block based on identifying that the transmission circuit operates; or turn off the signal detection block based on

identifying that a frequency of the transmission circuit is locked.

12. A method for operating an electronic device, the method comprising: detecting a signal output from a first VCO included in an RF circuit of the electronic device through a signal detection block included in the RF circuit; identifying whether the detected signal is an abnormal signal based on identifying a strength of the detected signal when the strength of the detected signal is less than a reference value; and controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the detected signal is abnormal.

13. The method of claim 12, further comprising: controlling at least one frequency divider included in a PLL of the second VCO to change a reference clock of the second VCO based on identifying that the identified signal is abnormal; and controlling the RF circuit to change an operating frequency of the second VCO based on identifying that a strength of a signal received through a reception circuit included in the RF circuit is less than a reference value.

14. The method of claim 12, further comprising: identifying a lookup table associated with an operating frequency combination of the electronic device based on identifying that the identified signal is abnormal; controlling the RF circuit to change the operating frequency of the second VCO based on identifying that the operating frequency combination is a first frequency combination; and controlling at least one frequency divider included in a PLL of the second VCO to change a reference clock of the second VCO based on at least one of: identifying that the operating frequency combination is a second frequency combination; or identifying that the operating frequency combination is not the first frequency combination.

15. A computer-readable storage medium recorded with instructions, when executed by at least one processor of an electronic device, cause the electronic device to perform at least one operation comprising: detecting a signal output from a first VCO included in an RF circuit of the electronic device through a signal detection block included in the RF circuit; identifying whether the detected signal is an abnormal signal based on identifying a strength of the detected signal when the strength of the detected signal is less than a reference value; and controlling the RF circuit to change an operating frequency of a second VCO included in the RF circuit based on identifying that the detected signal is abnormal.
